journey. “Ticks” on the light clock may be thought of as occurring every time the photon completes a round-trip—a billion ticks means that one second has elapsed.

We can use the light clock like a stopwatch to measure the time elapsed between events: We simply count how many ticks occur during the period of interest and multiply by the time corresponding to one tick. For instance, if we are timing a horse race and count that between the start and finish the number of round-trip photon journeys is 55 billion, we can conclude that the race took 55 seconds.

The reason we use the light clock in our discussion is that its mechanical simplicity pares away extraneous details and therefore provides us with the clearest insight into how motion affects the passage of time. To see this, imagine that we are idly watching the passage of time by looking at a ticking light clock placed on a nearby table. Then, all of sudden, a second light clock slides by on the table, moving at constant velocity (see Figure 2.2) The question we ask is whether the moving light clock will tick at the same rate as the stationary light clock?

To answer the question, let’s consider the path, from our perspective, that the photon in the sliding clock must take in order for it to result in a tick. The photon starts at the base of the sliding clock, as in Figure 2.2, and first travels to the upper mirror. Since, from our perspective, the clock is moving, the photon must travel at an angle, as shown in Figure 2.3. If the photon did not travel along this path, it would miss the upper mirror and fly off into space. As the sliding clock has every right to claim that it’s stationary and everything else is moving, we know that the photon will hit the upper mirror and hence the path we have drawn is correct. The photon bounces off the upper mirror and again travels a diagonal path to hit the lower mirror, and the sliding clock ticks. The simple but essential point is that the double diagonal path that we see the photon traverse is longer than the straight up-and-down path taken by the photon in the stationary clock; in addition to traversing the up-and-down distance, the photon in the sliding clock must also travel to the right, from our perspective. Moreover, the constancy of the speed of light tells us that the sliding clock’s photon travels at exactly the same speed as the stationary clock’s photon. But since it must travel farther to achieve one tick it